

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • About 10% of land is already cultivated and 25% grazed (1) • Most remaining land is too dry, wet, cold, hot or steep, is covered in ice, or has no soil (1)	2
01.2	<i>1 mark for each point.</i> • Land becomes too expensive for local people to rent / is used to grow export crops for more affluent societies (1) • So people with less money lose access to the land and food they used to produce (1)	2
01.3	<i>1 mark for each point.</i> • It uses natural processes to supply nutrients (legumes, recycled organic matter, soil biota), control pests and provide ecological services (1) • So that pesticides and artificial fertilisers become unnecessary rather than just being banned (1)	2

Question	Marking guidance	Marks
01.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Unsustainable features: reliance on chemical pesticides (resistance, non-target impacts, bans), non-renewable rock phosphate, fossil-fuel-intensive nitrate manufacture, over-exploitation of rivers / groundwater, salinisation, loss of soil organic matter, greenhouse gases from fuel, rice and livestock, loss of species providing ecosystem services • Pest control: cultural methods (weeding, mulching, crop rotation, barrier crops, culling), biological control, predator habitats, polyculture / companion crops, integrated control, reduced antibiotic use • Nutrients: recycling organic matter, crop rotation, cultivation of legumes, conservation of soil biota, reduced artificial fertiliser especially nitrate, low tillage • Water: low water-use crops, reservoirs and aquifer recharge, drip irrigation rather than sprays • Soil and carbon: low-tillage farming to maintain organic matter, reduced machinery, renewable energy, drought-tolerant rice varieties so fields are drained earlier, high-carbohydrate ground feed for cattle reduces methane • Biodiversity: conserve habitats of crop wild relatives, seed banks, retain hedgerows, ditches, ponds, woodland; agri-environment schemes • Evaluation: growing population (perhaps 12 billion) needs more food; lower trophic level diets; obstacles - cost, need for support, gaining acceptance of new techniques; learning from past land degradation (fertile crescent, US mid-west) • Judgement: no single strategy sufficient; combination of ecological principles and new technology; justified conclusion	9
	Model answer - Level 3 (8 marks)	

Question	Marking guidance	Marks
	Present agriculture relies on inputs that cannot last: chemical pesticides to which pests evolve resistance and which harm non-target species, rock phosphate that is non-renewable, nitrate fertiliser made with fossil-fuel energy, and irrigation that over-draws rivers and groundwater and salinises soil, while ploughing loses soil organic matter and livestock, rice and machinery release greenhouse gases. Each can be replaced by strategies based on ecological principles. Pests are controlled culturally by weeding, mulching, rotation, barrier crops and culling, by biological control and predator habitats, by polyculture and companion crops, and by integrated control that uses chemicals only as a last resort. Nutrients come from recycled organic matter, rotation with legumes and conserved soil biota, so artificial fertiliser, especially nitrate, can be reduced. Water is saved with low water-use crops, reservoirs and aquifer recharge and drip irrigation; soil carbon is kept by low tillage and less machinery, with renewable energy for what remains; methane falls with rice varieties that allow earlier drainage and with ground, carbohydrate-rich cattle feed. Habitats of crop wild relatives, seed banks, hedgerows, ditches, ponds and woodland and agri-environment schemes protect the biodiversity that supplies these services.	
	Food demand will rise with a population that may reach 12 billion, there is little new land, and imports can reduce food availability where it is grown, so yield cannot be ignored. Some strategies cost more or need farmers to accept new techniques, and past land degradation from the fertile crescent to the American mid-west shows what happens when they are not adopted. No single strategy suffices; a combination of ecological principles with new technology is the only route to feeding more people without exhausting the resources food depends on.	
	Weaker answer - Level 1 (2 marks)	
	To make agriculture sustainable farmers should use fewer chemicals. Organic farming does not use pesticides or artificial fertilisers. Crop rotation and manure keep the soil fertile. Farmers should not waste water. Hedges should be kept for wildlife. This would make farming more sustainable but it might produce less food.	
	What makes the difference	
	• Level 1 gives general advice; Level 3 identifies each unsustainable feature and matches it with specific strategies, showing understanding of why they work.	
	• Level 3's evaluation considers population growth, land limits, cost, adoption and historical evidence; Level 1's is a single clause.	
	• Note that Level 3 avoids defining organic farming as 'not using chemicals' - the textbook stresses it is the positive use of natural processes.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	
02.1	<i>Correct answer only.</i> • B (1)	1
02.2	<i>1 mark for each point.</i> • Number all possible sampling sites (eg on a grid map) (1) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers) (1)	2

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> • Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis (1) • So the sample does not represent the whole area and conclusions are invalid (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
03.1	<i>1 mark for each point.</i> • In-situ: conserving species in their natural habitat with the whole community; ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding (1) • Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching) (1)	2
03.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Is the wild population threatened? (1) • Is there a genetically diverse captive population? (1) • Is in-situ conservation failing? (1) • Is keeping a captive population realistic (habitat size, food, relationships, cost)? (1) • Is release likely to be successful now or in future (suitable habitat, threats removed)? (1)	4
03.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Blue whale: habitat size far too large to provide; food (huge quantities of krill / plankton) cannot be supplied; cost (2) • Large blue butterfly: complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs; requires specific ant species and thyme food plant (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3.	
04.1	<i>Correct answer only.</i> • B (1)	1
04.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Harvesting no faster than the MSY - resource never depleted (1) • Mixed species plantations - more food choices, relationships and niches (1) • Indigenous species - support indigenous insects, birds (1) • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield (1) • Selective logging - small clearings, less disturbance, long-term timber benefit (1) • Retention of dead wood, buffer zones, corridors (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	

Question	Marking guidance	Marks
05.1	<i>1 mark for each point.</i> - Isolation makes colonisation difficult: only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them (1) - Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists (1) - Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii) (1) - Endemics have small ranges and populations so are vulnerable to extinction (1)	4
05.2	<i>1 mark for each point.</i> - The birds evolved without mammal predators so nest on the ground / in burrows with no defence (1) - Rats eat eggs and chicks; numbers fell to very low levels (45 and 12 pairs) (1) - After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor (1)	3
05.3	<i>1 mark for each point.</i> - Introduced mosquitoes carry avian malaria to which endemic birds have no resistance; birds survive only above about 1500 m where it is too cold for malaria (1) - Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range (1)	2
05.4	<i>1 mark.</i> Control of developments and visitor numbers (biosecurity, quarantine of imports); protected areas; captive breeding of endemics (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4.	
06.1	<i>1 mark for each point.</i> - Anchovy r-selected; blue whale k-selected (1) - r-selected: mature quickly, produce many young, disperse widely, short lives; adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) (1) - k-selected: mature late, few young with parental care, long-lived; populations stable near carrying capacity (whales, elephants, rhinos) (1)	3
06.2	<i>1 mark for each point.</i> - Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly (1) - An increase in death rate above the low birth rate causes a population crash (1) - Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium); r-selected species rebound within a few years once exploitation stops (1)	3
06.3	<i>1 mark.</i> New population = starting population + births + immigrants - deaths - emigrants (1)	1

Question	Marking guidance	Marks
06.4	<i>1 mark for each point.</i> • The greatest rate of exploitation (harvest) possible without causing long-term population decline (1) • Equals the surplus production: highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) (1) • At carrying capacity net growth is zero so any harvest reduces the population; harvesting above MSY causes decline (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	
07.1	<i>1 mark for each point.</i> • All greenhouse gases expressed as CO₂ equivalent (allowing for different warming strengths) (1) • Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) (1) • A fair share of the infrastructure: roads, car parks, signals, street lighting (1)	3
07.2	<i>1 mark for each point.</i> • Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels (1) • Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time (1)	2
07.3	<i>1 mark for each point.</i> • Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too (1) • Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth (1) • Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated (1) • Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected (1) • Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers (1)	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	
08.1	<i>1 mark for each point.</i> • Food: cod, tuna, swordfish, sharks, turtles; extinct - dodo, great auk, passenger pigeon (1) • Fashion: leopard, snow leopard, tiger, ocelot fur; fur seals; crocodile / alligator leather; kingfisher, parrot, ostrich feathers (1) • Entertainment / pets: parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants (1) • Furniture / ornaments: mahogany, teak, ramin timber; ebony and ivory piano keys; turtle shell, shark teeth, coral and shells as souvenirs (1) • Traditional medicines: tiger (bones, claws), rhinoceros horn, seahorses (1)	5

Question	Marking guidance	Marks
08.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Threat to humans: sharks, venomous snakes, crocodiles (1) • Disease vectors: malaria mosquitoes, tsetse flies (1) • Predators of livestock: wolves, puma, birds of prey, herons, seals (1) • Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers (1)	3
08.3	<i>1 mark for each point.</i> • High demand and prices make illegal trade / poaching profitable; laws are hard to enforce across borders (1) • Belief in the medicine persists despite little scientific evidence; species become rarer and more valuable (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	
09.1	<i>Correct answer only.</i> • Waste products that are the raw materials for other processes (1)	1
09.2	<i>1 mark for each point.</i> • Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) (1) • The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂ (1)	2
09.3	<i>1 mark for each point.</i> • Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials (1) • Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
10.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(15 - 6) / 6 \times 100$ (1) • = 150% (1)	2
10.2	<i>1 mark for each point.</i> • Cells shed by the newts into the water contain their DNA (1) • Water samples are analysed for DNA sequences unique to the species (1)	2
10.3	<i>1 mark for each point.</i> • Catches depend on trapping effort, weather and newt activity on the night, so they vary from year to year (1) • Trapping only samples one pond and one part of the population, whereas eDNA surveys all 20 ponds with a standard method (1)	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Image databases of unique markings (tiger stripes, whale shark spots, dolphin fins) giving territory size, movements, lifespan and social groups; limited to species with unique features • Motion-sensitive and infrared cameras, CCTV of nests with less disturbance; marking (rings, wing tags) • DNA: eDNA to detect presence (great crested newts, invasive carp); DNA databases to identify regional populations and trace timber, ivory and fish • Auditory monitoring: bat detectors and sonograms for presence, abundance and activity • Position monitoring: radio, GPS and acoustic transmitters; geolocator tags for small birds; data recorders; carrier systems - ROVs / drones, AUVs, balloons, aircraft, satellites, animals • Satellites: visible light for land use and deforestation, infrared for vegetation and temperature, radar for wind, waves, sea level and oil; GPS for tracking • Evaluation: greater coverage, less disturbance and continuous data vs cost, battery / weight limits, calibration, need for ground-truthing, ethical issues of tagging; indirect evidence and traditional methods still needed • Justified conclusion	9
	Model answer - Level 3 (9 marks)	
	New technologies have transformed what can be monitored and how often. Image databases of unique markings - tiger stripes, whale-shark spots, dolphin fin scars - let individuals be recognised, giving territory size, movements, lifespan and social structure without repeated capture, though only for species with unique features. Motion-sensitive and infrared cameras and CCTV record activity day and night with far less disturbance than observers, especially at nests.	

Question	Marking guidance	Marks
	Genetics adds detection where nothing can be seen: eDNA in water reveals great crested newts or invasive carp from shed cells, and DNA databases identify which regional population timber, ivory or fish came from, supporting enforcement. Bat detectors and sonograms give presence, abundance and activity of species heard but not seen. Position monitoring uses radio, GPS and acoustic transmitters, or tiny geolocator tags for birds too small to carry a transmitter, while data recorders on drones, AUVs, balloons, aircraft, satellites or the animals themselves collect abiotic data. Satellites map land use and deforestation in visible light, vegetation density and temperature in infrared, and wind, waves, sea level and oil slicks with radar that works through cloud.	
	The gains are coverage, continuity and reduced disturbance. The limits are cost, battery life and weight, the need to calibrate sensors and ground-truth remote data, ethical questions about tagging, and the fact that many species still require traditional methods and indirect evidence. New technology therefore complements rather than replaces fieldwork, but for rare, remote or nocturnal species it has made effective conservation possible where it was not before.	
	Weaker answer - Level 1 (3 marks)	
	New technologies help monitor wildlife. Cameras can take pictures of animals at night. GPS collars can track where animals go. Satellites can see deforestation from space. DNA can tell which animals are present. Bat detectors can hear bats. These help conservation because we know where animals are and how many there are. They can be expensive though.	
	What makes the difference	
	• Level 1 lists technologies with no detail of how they work or what data they give; Level 3 links each to the information it provides (territory size, presence, activity).	
	• Level 3's evaluation considers several limitations (calibration, ground-truthing, ethics, species coverage) and reaches a balanced judgement; Level 1 has one word - 'expensive'.	
	• Structure: Level 3 groups by imagery, genetics, sound, position and remote sensing; the list in Level 1 has no organisation.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	

Question	Marking guidance	Marks
11.1	Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations; positive feedback and tipping points caused by humans • Energy: solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process); renewable energy, heat pumps, bioleaching, fuel cells • Materials: cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals; recycling, Cradle to Cradle, industrial symbiosis • Diversity and stability: rainforests / reefs vs agroecosystems; diverse energy mixes • Optimum vs maximum production; supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics; limits and judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	25

Question	Marking guidance	Marks
11.2	Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Marine productivity: light, nutrients, upwelling; where fisheries occur • Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	25